

ANN LIA SAM



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National Institute of Science Education and Research, Bhubaneswar, Odisha

RESEARCH INTERESTS

I am a 2nd year Integrated M.Sc. student pursuing Physics from NISER, Bhubaneswar. My current Research Interest lies in Quantum Information Processing, Quantum Computation, Light Matter Interactions, specifically in Cavity QED. I am currently seeking to apply experimental insights from strong coupling to the theoretical model of decoherence and quantum logic gates.

EDUCATION

National Institute of Science Education and Research, Bhubaneswar	2024 - Present
- Integrated M.Sc. - Mathematics, Biology, Physics, Chemistry - CGPA - 7.38/10	
St. Antony's Public School, Kottayam, Kerala	2021-2023
- Central Board of Secondary Education, 12th - Mathematics, Biology, Physics, Chemistry - CGPA - 9.6/10	
Holy Trinity Vidyabhavan, Karthikappally, Kerala	2021
10th - CGPA - 10/10	

ACHIEVEMENTS

- Secured AIR - 397 in National Entrance Screening Test (NEST) 2024
- Secured AIR - 3148 in IISER Aptitude Test (IAT) 2024
- Recipient of DAE - DISHA scholarship.

INTERSHIPS AND RESEARCH EXPERIENCE

- Summer Research Intern | Prof. Anoop Thomas Lab, IISc Bangalore (May – July 2024)
 - Investigated Vibrational Strong Coupling (VSC) and its effect on Photoinduced Electron Transfer.
 - Fabricated Fabry-Pérot microcavities using spin-coating and measured transmission spectra.
 - Analyzed biexponential fluorescence decay kinetics using Time-Resolved Spectroscopy.
 - Studied the hybridization of molecular vibrations with cavity vacuum fields.

TECHNICAL SKILLS

- **Experimental:** Precision Optical Alignment, Spin-coating, FTIR Spectroscopy, UV-Vis/Fluorescence Spectroscopy, Michelson & Fabry-Pérot Interferometry.
- **Computational:** Python (NumPy, Matplotlib, SciPy)
- **Instruments:** Handled spectrophotometers and laser-based lab setups.

PROJECTS & LAB WORK

- *Linear Optics Lab (Ongoing): Studying Gaussian beam propagation, beam waist measurement, and interference patterns.*

ADDITIONAL INFORMATION

- **Languages known:** English, Malayalam, Tamil (intermediate), Hindi (intermediate).
- **Hobbies:** Reading, Singing, Dancing, Playing Violin
- Active Member in Coding Club